

timmy

Project Report

1. Readme

Timmy clone

This is a clone of the Paul Cochrane Timmy overdrive.

This pedal is verified, though the layout is poor, IMHO.

It is the first pedal I did in Kicad, and it took 4 revisions to get a working pedal, going all the way back to 2022. This was in the hall of shame box for some time.

v1: pads for switch too small. Changed to 2.5mm hole

v2: RV2 is backwards from the tagboardeffects layout. I.e. the schematic has multiple issues.

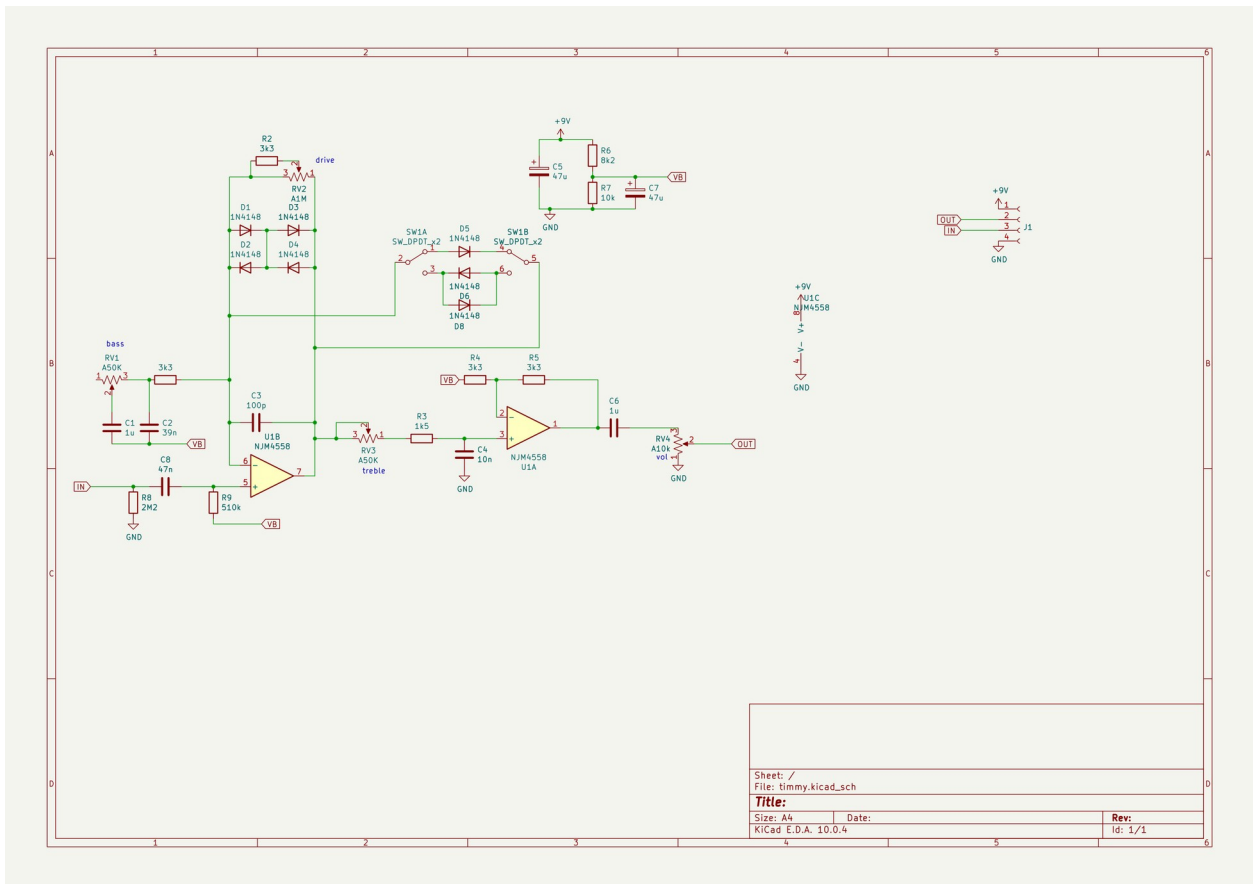
Pedal has low volume. Managed to get this to work correctly via a series of bodes. Also note te 3.3K resistor R2 is diff in tagboardeffects layout

v3: Seems to work correctly. Missing some values on the silkscreen layer.

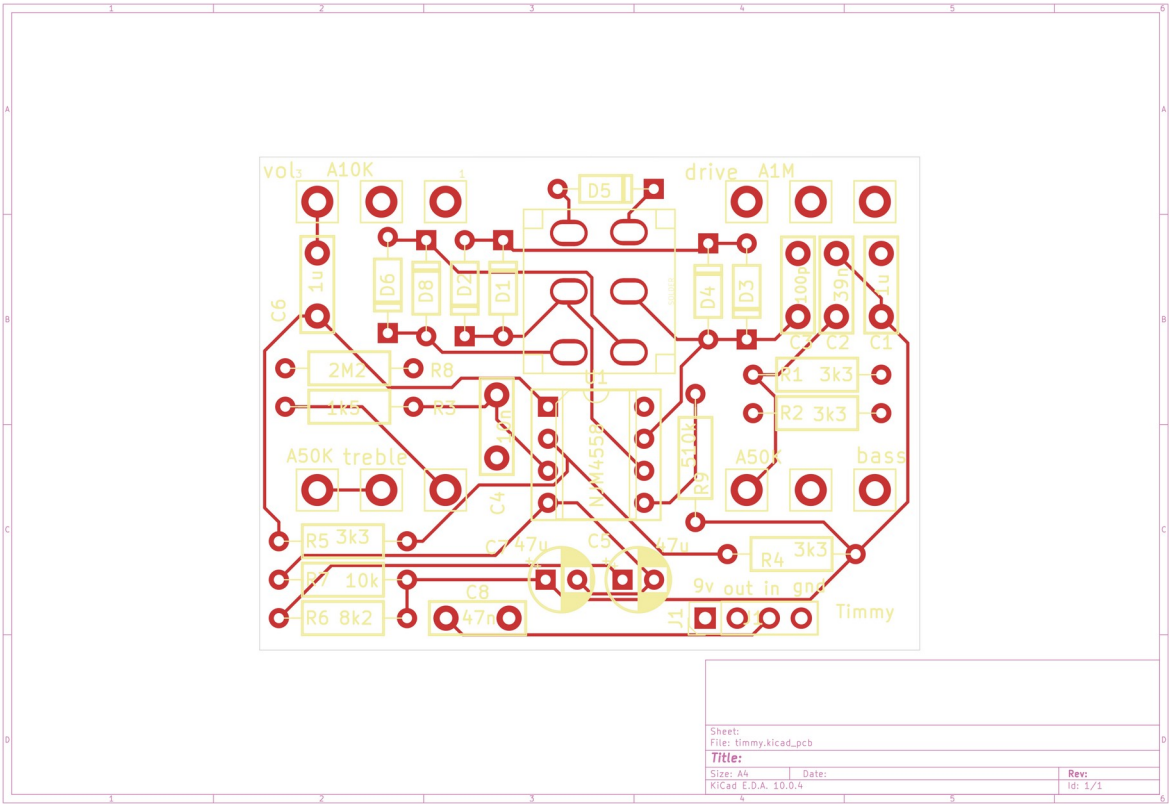
v4: final version.

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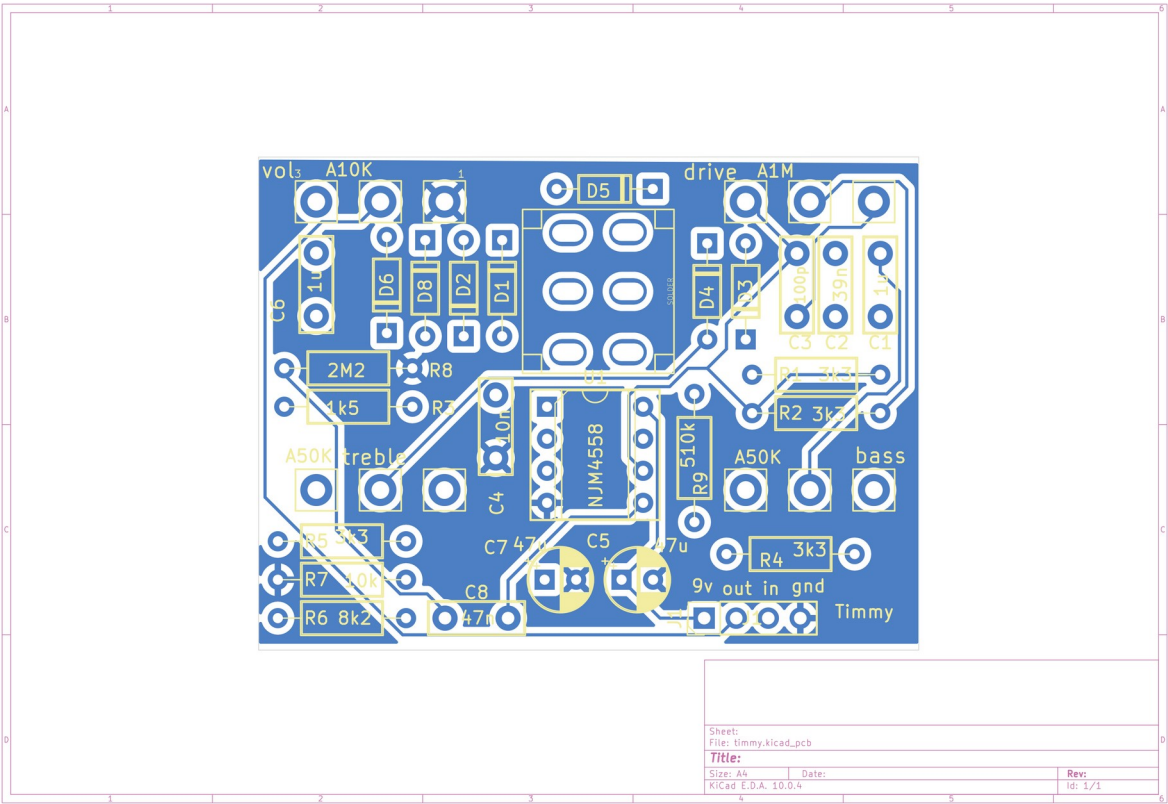
2. Schematic



3. PCB F.Cu + Silkscreen



4. PCB B.CU + Silkscreen



5. Bill of Materials

Reference	Value	Footprint	Qty	DNP
C1,C6	1u	Capacitor_THT: C_Disc_D7.5mm_W2.5mm_P5.00mm	2	
C2	39n	Capacitor_THT: C_Disc_D7.5mm_W2.5mm_P5.00mm	1	
C3	100p	Capacitor_THT: C_Disc_D7.5mm_W2.5mm_P5.00mm	1	
C4	10n	Capacitor_THT: C_Disc_D7.5mm_W2.5mm_P5.00mm	1	
C5,C7	47u	Capacitor_THT: CP_Radial_D5.0mm_P2.50mm	2	
C8	47n	Capacitor_THT: C_Disc_D7.5mm_W2.5mm_P5.00mm	1	
D1-D6,D8	1N4148	Diode_THT:D_DO-35_SOD27_P7.62mm_Horizontal	7	
J1	Conn_01x04_Female	Connector_PinHeader_2.54mm:PinHeader_1x04_P2.54mm_Vertical	1	
R1,R2,R4,R5	3k3	Resistor_THT:R_Axial_DIN07_L6.3mm_D2.5mm_P10.16mm_Horizon	4	

		tal		
R3	1k5	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal	1	
R6	8k2	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal	1	
R7	10k	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal	1	
R8	2M2	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal	1	
R9	510k	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal	1	
RV1,RV3	A50K	pedals:16MM_ PCB_PADS	2	
RV2	A1M	pedals:16MM_ PCB_PADS	1	
RV4	A10k	pedals:16MM_ PCB_PADS	1	
SW1	SW_DPD_T_x2	switches:DPD T.MINI3	1	
U1	NJM4558	Package_DIP: DIP- 8_W7.62mm_S	1	

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Total line items: 19